

SOA Survey

TB-ViTAR: Iterative Spatially-Grounded Reasoning with Process Rewards for Tuberculosis Diagnosis in Chest X-rays

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Abstract

Medical vision-language models (VLMs) trained with reinforcement learning have shown strong diagnostic reasoning on radiology benchmarks. However, they process images in a single forward pass without iteratively focusing on suspicious regions, their reward functions only check whether the final answer is correct while ignoring whether intermediate reasoning steps are spatially or clinically valid, and none of them have been adapted for a specific disease like tuberculosis. This survey reviews 14 papers across three research fronts: (i) GRPO-based medical VLM reasoning, (ii) spatially-grounded reward design for medical image grounding, and (iii) iterative visual reasoning frameworks that model clinical diagnostic workflows. We identify a gap at their intersection: no existing work combines the iterative Think-Act-Rethink-Answer reasoning paradigm with per-step spatially-grounded process rewards for disease-specific diagnosis. We propose TB-ViTAR, a system that adapts this combined framework to tuberculosis CXR diagnosis on TBX11K, where bounding-box annotations enable precise spatial verification of each reasoning step.

1. Problem Definition

Tuberculosis kills 1.3 million people every year, most of them in countries where there are not enough radiologists to read the chest X-rays that could catch the disease early. Deep learning classifiers have reached above 95% AUC on controlled TB benchmarks [12, 13], but they output a binary label with no explanation of what abnormality was found, where it is, or why it suggests TB.

Vision-language models trained with GRPO can generate natural-language reasoning alongside their answers. MedVLM-R1 [1] showed that a 2B model trained with GRPO on 600 samples outperforms mod-

els 36 times its size. But current medical VLMs have three problems that prevent them from matching how a radiologist actually works.

First, they reason in a **single pass**. A radiologist does not look at an X-ray once and produce a diagnosis. They scan the full image, identify a suspicious region, zoom in on it, examine it closely, and only then conclude. Current VLMs skip this iterative focus entirely.

Second, their reward functions are **flat**. They score the final answer as correct or not. A model that says “Yes, TB” after reasoning through the right findings and one that says “Yes, TB” by accident get the same reward. The quality of each reasoning step is invisible to the training signal.

Third, they are **generic**. Every GRPO-trained medical VLM works on general radiology VQA. None has been adapted to a specific disease like TB, where particular visual patterns (cavities, upper-lobe predominance, tree-in-bud nodularity) and particular clinical stakes (missed TB cases are lethal) call for specialized training.

Recent work has begun to address the first two problems separately. ViTAR [5] introduced a Think-Act-Rethink-Answer loop for iterative medical reasoning. MedGround-R1 [4] designed spatial-semantic rewards for GRPO-based medical image grounding. ChestX-Reasoner [6] proposed process rewards that evaluate each reasoning step individually. But nobody has combined iterative reasoning with per-step spatial process rewards, and nobody has applied any of these techniques to a specific disease.

2. Literature Review

2.1. Challenge 1: GRPO for Medical VLMs — Accurate but Single-Pass with Flat Rewards

Group Relative Policy Optimization, introduced in DeepSeek-R1 [3], trains language models by generating multiple candidate responses per prompt, scoring them, and updating the policy to favor above-average candidates. Unlike PPO, GRPO needs no critic network, cutting compute roughly in half.

MedVLM-R1 [1] applied GRPO to medical VQA by training Qwen2-VL-2B on 600 MRI samples. Accuracy went from 55.1% to 78.2% across MRI, CT, and X-ray, beating Qwen2-VL-72B. The model also began generating natural-language reasoning on its own, without being given any reasoning examples. Med-R1 [2] extended this across 8 imaging modalities, achieving a 29.9% gain over base models. A notable finding: their “No-Think” variant, which skips intermediate reasoning, sometimes performed better than the thinking version, suggesting that poorly calibrated reasoning hurts more than it helps. MediX-R1 [9] (a recent preprint) introduced composite rewards combining LLM-based accuracy with medical semantic signals for open-ended generation.

All three papers share two limitations. They process images in one forward pass without revisiting specific regions, and their rewards only evaluate the final answer. The reasoning quality itself — whether the model identified the right anatomy, described the right finding, and located it correctly — is never checked during training.

2.2. Challenge 2: Spatial Grounding via GRPO — Works but Only on the Final Output

MedGround-R1 [4] (MICCAI 2025) tackled spatial grounding for medical VLMs. They designed two reward signals: a spatial reward based on IoU between predicted and ground-truth bounding boxes, and a semantic reward using frozen MedCLIP to measure alignment between the predicted region and the referring text. They also introduced the Chain-of-Box template: during the model’s thinking phase, the model writes bounding-box coordinates alongside its text reasoning, linking spatial claims to actual image regions. On MS-CXR and ChestX-ray8, this reached

state-of-the-art grounding accuracy. Removing the spatial reward caused a severe drop in performance, confirming that spatial signals are needed.

VALOR [10] applied a two-stage GRPO approach to radiology report generation: Stage 1 used textual rewards to improve clinical terminology, Stage 2 aligned the visual encoder with disease findings through spatial signals. MedMO combined GIoU-based bounding-box rewards with factuality checks in a three-stage pipeline spanning multiple medical domains.

The shared limitation: all three evaluate spatial grounding only on the model’s final output. None of them checks whether intermediate reasoning steps (e.g., “I observe haziness in the right upper lobe” before concluding “consistent with TB”) are themselves spatially valid. A model can produce correct final coordinates while its reasoning chain contains spatially inaccurate claims. This disconnect between reasoning quality and output quality is not addressed by any current grounding work.

2.3. Challenge 3: Iterative Reasoning — Clinically Motivated but Missing Spatial Process Verification

Radiologists do not diagnose in one pass. They scan the full image, focus on suspicious areas, examine details, and then form a conclusion. ViTAR [5] modeled this clinical workflow as a four-step cognitive chain:

1. **Think:** observe the full image, generate an initial assessment
2. **Act:** identify key regions by drawing bounding boxes on the image
3. **Rethink:** examine those regions, refine the assessment with additional detail
4. **Answer:** produce the final diagnosis

Trained with SFT on 1K expert-crafted examples followed by GRPO on 16K VQA samples, ViTAR outperformed both Med-R1 and MedVLM-R1 across multiple benchmarks. Visual attention analysis showed that between the Think and Rethink rounds, the model’s attention shifted to tighter, more clinically relevant regions, confirming genuine iterative

refocusing rather than trivial repetition.

Med-VRAgent [7] (EMNLP 2025) took a tree-search approach: Monte Carlo Tree Search explores multiple reasoning paths, a Teacher agent provides ROI-specific visual guidance, and trajectories are used to fine-tune with PPO. MedVCTP [8] (NeurIPS 2025) proposed a training-free See-Think-Confirm pipeline where visual features are extracted, interpreted through chain-of-thought prompting, and verified through cross-modal CLIP similarity.

The shared limitation across all three: iterative reasoning is supervised only through outcome rewards on the final answer. ViTAR’s GRPO does not check whether the Think step identified the correct lung region, the Act step drew accurate bounding boxes, or the Rethink step added valid clinical details. Each intermediate step could be wrong and the model could still stumble onto the right final answer. ChestX-Reasoner [6] partially addressed this by introducing process factuality rewards that verify each reasoning step against clinical reports, but it uses single-pass reasoning (not iterative) and does not perform spatial grounding.

The gap is clear: nobody has combined ViTAR-style iterative reasoning with MedGround-R1-style spatial verification at every step, and ChestX-Reasoner-style process rewards that evaluate reasoning quality step by step.

Open Problems (flagged by the field):

- ViTAR [5] notes that “the mechanisms underlying iterative visual cue processing remain insufficiently understood” and that per-step supervision is needed to improve reliability.
- MedGround-R1 [4] reports that “removing the spatial reward causes performance collapse,” confirming that spatial signals are critical but acknowledging they only evaluate the final output.
- ChestX-Reasoner [6] finds that process rewards improve reasoning factuality, but their approach is limited to single-pass reasoning without spatial grounding.
- MedVLM-R1 [1] calls for domain-specific GRPO applications beyond general radiology VQA.
- MedRAX [14] explicitly lists “applying reinforcement learning to boost reasoning ability” as future work for CXR agents.

3. Research Gap and Proposed Direction

Three validated techniques exist independently: (1) GRPO with spatial rewards for medical image grounding (MedGround-R1), (2) iterative Think-Act-Rethink-Answer reasoning for medical VLMs (ViTAR), and (3) per-step process rewards for reasoning verification (ChestX-Reasoner). No existing paper combines more than one of these three ideas. The intersection of all three — iterative reasoning where each step is verified by a spatially-grounded process reward — is completely unexplored. On top of this, no current medical VLM with GRPO has been adapted for a specific disease.

We propose **TB-ViTar**: an iterative reasoning framework for TB CXR diagnosis that combines the Think-Act-Rethink-Answer structure from ViTAR with per-step spatially-grounded process rewards. TBX11K [11] provides the bounding-box annotations that make this possible: we can verify at each reasoning step whether the model’s spatial claims match the ground-truth TB lesion locations.

The process rewards work as follows. At the Think step, we check whether the model mentioned the correct lung zone (spatial accuracy, verified via bounding-box coordinates). At the Act step, we check whether the region the model selected overlaps with the ground-truth TB annotation (IoU reward). At the Rethink step, we check whether the refined assessment includes clinically specific TB features like cavity, consolidation, or infiltrate (clinical reward). At the Answer step, we check overall diagnostic accuracy with an asymmetric penalty for false negatives, since missed TB cases are more dangerous than false alarms in a screening setting. We propose a two-step Think-Answer pipeline where the model generates spatially-grounded reasoning with Chain-of-Box coordinates, trained with GRPO using per-step spatial and clinical process rewards on TBX11K.

Research Question (formal): Can a compact VLM (Qwen2-VL-2B), trained with GRPO using per-step

spatially-grounded process rewards within an iterative Think-Act-Rethink-Answer reasoning framework, outperform single-pass outcome-rewarded baselines on both diagnostic accuracy and reasoning quality for TB chest X-ray interpretation?

4. Benchmark, Evaluation, and Data Landscape

Table 1: Key datasets.

Dataset	Size	Type	Notes
TBX11K	11.2K	CXR	TB bboxes, 4 classes; primary
MS-CXR	1.1K	CXR	Phrase grounding; eval for spatial accuracy
Shenzhen	662	CXR	TB/normal; cross-dataset test
Montgomery	138	CXR	TB/normal + masks; cross-dataset test
SLAKE	14K	Mixed	Medical VQA; general eval
VQA-RAD	3.5K	Mixed	Radiology VQA; general eval

Datasets. TBX11K provides approximately 11,200 CXRs with bounding-box annotations for TB lesions. These boxes are essential for two purposes: generating VQA training pairs with spatial information, and computing per-step process rewards (IoU between the model’s claimed coordinates and the ground-truth). MS-CXR provides phrase-grounding evaluation. Shenzhen and Montgomery test cross-dataset generalization to unseen hospital distributions.

Evaluation Protocol. Four levels: (1) **Classification:** sensitivity, specificity, AUC-ROC vs. CNN baselines. (2) **VQA accuracy:** exact-match and LLM-as-judge scoring on TB questions. (3) **Spatial grounding:** IoU between the coordinates in the model’s reasoning chain and ground-truth bounding boxes, following MedGround-R1’s protocol. (4) **Per-step reasoning quality:** manual evaluation of 100 samples using a rubric adapted from ChestX-Reasoner’s

RadRScore, scoring factuality, completeness, and clinical specificity at each reasoning step separately.

Compute Context. ViTAR trained Qwen2-VL-7B with SFT+GRPO on 17K samples. MedGround-R1 trained with GRPO in 5K steps on a single GPU. We use the 2B model on Colab Pro A100 40GB for the two-step Think-Answer pipeline (estimated 8–12 hours).

Experimental Design Considerations

- **Baselines:** (1) ResNet-50 on TBX11K (CNN accuracy ceiling). (2) Qwen2-VL-2B zero-shot (untrained VLM). (3) Qwen2-VL-2B + SFT on same TB VQA data (isolates RL contribution). (4) Qwen2-VL-2B + outcome-only GRPO (MedVLM-R1 style — isolates process reward contribution). (5) MedVLM-R1 checkpoint zero-shot on TB (tests generic GRPO transfer). The two critical comparisons: outcome-only vs. process-reward GRPO (tests reward contribution), and single-pass vs. iterative reasoning (tests architectural contribution).
- **Ablations:** (1) *Reward decomposition:* outcome-only → outcome+spatial → outcome+spatial+clinical process rewards, to measure the contribution of each reward component. (2) *Reasoning depth:* Think-Answer (2-step) vs. Think-Act-Rethink-Answer (4-step). (3) *Data efficiency:* 200, 500, 1000, 2000 VQA pairs. (4) *Cross-dataset:* TBX11K → Shenzhen/Montgomery.
- **Pitfalls:** TBX11K has class imbalance (~924 active TB vs. ~5,000 healthy); use stratified splits and AUC-PR alongside AUC-ROC. Template VQA pairs could create artifacts; include manually written held-out questions. Per-step process rewards may cause reward hacking — monitor for reward inflation without accuracy gains. Cross-dataset eval on Shenzhen/Montgomery guards against overfitting to TBX11K distributions.

5. Annotated Reading List

References

- [1] J. Pan et al. MedVLM-R1: Incentivizing Medical Reasoning Capability of VLMs via Reinforcement Learning. *MICCAI*, 2025. *What: GRPO on Qwen2-VL-2B with 600 MRI samples; 2B model beats 72B models; 35% X-ray OOD gain over SFT. How: group-relative advantage computation with accuracy+format reward; no critic network. Why here: foundational GRPO method and our starting codebase; we extend from outcome-only to per-step process rewards.*
- [2] Y. Lai et al. Med-R1: Reinforcement Learning for Generalizable Medical Reasoning in VLMs. *arXiv:2503.13939*, 2025. *What: GRPO across 8 modalities and 5 task types; Think vs. No-Think ablation. How: open-source training code with modality-specific configs and cross-modal evaluation. Why here: provides training infrastructure we build on; No-Think finding motivates our per-step verification.*
- [3] D. Guo et al. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via RL. *arXiv:2501.12948*, 2025. *What: pure RL produces reasoning without SFT using GRPO. How: eliminates critic by computing advantages relative to group of sampled responses. Why here: the foundational RL algorithm; understanding GRPO mechanics is essential for our reward design.*
- [4] H. Xu et al. MedGround-R1: Spatial-Semantic Rewarded GRPO for Medical Image Grounding. *MICCAI*, 2025. *What: first RL framework for medical image grounding using spatial IoU + MedCLIP semantic rewards. How: Chain-of-Box template embeds bbox coordinates inside the model’s thinking chain. Why here: our spatial reward design builds on their IoU reward; we extend Chain-of-Box from single-pass to iterative.*
- [5] K. Chen et al. ViTAR: Think Twice to See More — Iterative Visual Reasoning in Medical VLMs. *arXiv:2510.10052*, 2025. *What: Think-Act-Rethink-Answer iterative reasoning; attention tightens from Think to Rethink round. How: SFT on 1K expert examples + GRPO on 16K VQA samples with Qwen2-VL-7B. Why here: our core architectural blueprint; we add spatial process rewards that ViTAR lacks.*
- [6] Z. Fan et al. ChestX-Reasoner: Advancing Radiology Foundation Models with Step-by-Step Verification. *arXiv:2504.20930*, 2025. *What: process factuality rewards that verify each reasoning step against clinical reports. How: mines reasoning chains from reports; GRPO with process + outcome rewards on Qwen2-VL-7B. Why here: our process reward concept extends their per-step verification to include spatial grounding.*
- [7] G. Guo, X. Lu, Y. Feng. Med-VRAgent: Medical Visual Reasoning-Enhanced Agents. *EMNLP*, 2025. *What: MCTS + visual guidance + PPO for multi-path medical reasoning with Teacher-Student-Assessor. How: adaptive tree search explores reasoning paths; trajectories used for PPO fine-tuning. Why here: validates tree-based reasoning for medical VLMs; our iterative approach is simpler but combined with spatial verification.*
- [8] MedVCTP Authors. MedVCTP: Improving Accuracy and Explainability in Medical Visual Reasoning. *NeurIPS*, 2025. *What: See-Think-Confirm modular pipeline with CLIP-based cross-modal verification. How: training-free; regional captioning + CoT reasoning + verification loop. Why here: validates modular iterative reasoning; our approach trains this structure end-to-end with GRPO instead of using it inference-only.*
- [9] S. Mullappilly et al. MediX-R1: Open-Ended Medical Reinforcement Learning. *arXiv:2602.23363*, 2026. (Preprint.) *What: GRPO for free-form medical responses with composite LLM+semantic rewards across 16 modalities. How: single-stage RL with composite reward combining accuracy, format, and semantic signals. Why here: validates composite rewards for medical GRPO; our per-step process rewards extend this concept.*

- [10] S. Bose et al. VALOR: Visual Alignment via GRPO for Grounded Radiology Report Generation. *arXiv:2512.16201*, 2025. *What: two-stage GRPO — Stage 1 textual rewards, Stage 2 spatial alignment for report generation. How: trains vision projection module to attend to disease-relevant regions. Why here: validates staged spatial GRPO; our multi-step process rewards extend this beyond two stages.*
- [11] Y. Liu et al. Rethinking Computer-Aided Tuberculosis Diagnosis. *CVPR*, 2020. *What: TBX11K dataset — 11.2K CXRs with bounding-box annotations for TB lesions in PASCAL VOC format. How: 4 classes (healthy, sick-non-TB, active TB, latent TB); AUC loss for class imbalance. Why here: our primary dataset; bounding boxes enable both VQA pair generation and per-step spatial reward computation.*
- [12] E. Kotei and R. Thirunavukarasu. A Review on Deep Learning for TB Detection from CXR. *Arch. Comp. Methods Eng.*, 31, 455–474, 2024. *What: systematic review of all DL approaches for TB CXR detection. How: covers CNNs, transfer learning, augmentation; confirms 95%+ AUC on benchmarks. Why here: establishes that classification accuracy is solved but interpretability and localization are not.*
- [13] A. Mirugwe et al. Improving TB Detection in CXR Through Transfer Learning and Deep Learning. *JMIRx Med*, 6, e66029, 2025. *What: compares 6 CNN architectures (VGG, ResNet, Inception variants) for TB classification. How: VGG16 achieves best accuracy-efficiency trade-off at 96.7% on 4,200 CXRs. Why here: most recent CNN baseline; our model must match this accuracy while also providing grounded reasoning.*
- [14] A. Fallahpour et al. MedRAX: Medical Reasoning Agent for Chest X-ray. *ICML*, 2025. *What: ReAct-loop agent orchestrating 8 CXR specialist tools; 63.1% on ChestAgentBench. How: LangChain/LangGraph with modular tool selection; beats GPT-4o by 6.7 points. Why here: shows multi-step CXR reasoning is achievable; explicitly lists RL as future work direction.*

6. Preliminary Implementation Resources

- <https://github.com/Yuxiang-Lai117/Med-R1> — Med-R1 GRPO training code for Qwen2-VL. Primary training codebase.
- <https://huggingface.co/JZPeterPan/MedVLM-R1> — Pre-trained MedVLM-R1 weights. Baseline for zero-shot TB evaluation.
- <https://github.com/MAGIC-AI4Med/ChestX-Reasoner> — ChestX-Reasoner with process reward implementation. Reference for our per-step reward design.
- <https://mmcheng.net/tb/> — TBX11K dataset with bounding-box annotations.
- <https://github.com/EvolvingLMs-Lab/open-r1-multimodal> — Open-source multimodal GRPO training framework.
- <https://github.com/KwongFuk/Med-VRAgent> — Med-VRAgent code with MCTS and visual guidance. Reference for iterative reasoning.
- <https://github.com/bowang-lab/MedRAX> — MedRAX codebase. Reference for tool-based CXR reasoning.